Unit: 6 Unit 8: Quadratics **Name:** ______________
Topic: Graphing Quadratics and Properties of Parabolas **Date:** __________

Foundation (Jalapeno)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?
(A) $(0, 0)$
(B) $(2, 0)$
(C) $(-2, 0)$
(D) $(-2, 4)$
(E) $(0, 4)$

Q2. A spaceship is traveling through space with a velocity of $v = t^2 - 4t + 3$. What is the axis of symmetry of its velocity graph?
(A) $t = -1$
(B) $t = 1$
(C) $t = 2$
(D) $t = 3$
(E) $t = 4$

Q3. What is the maximum value of the function $y = -x^2 + 2x + 1$?
(A) $y = -1$
(B) $y = 0$
(C) $y = 1$
(D) $y = 2$
(E) $y = 3$

Q4. A planetary orbit can be modeled by the equation $y = x^2 - 6x + 9$. What is the direction of the parabola?
(A) Upward
(B) Downward
(C) Leftward
(D) Rightward
(E) Horizontal

Q5. What are the x-intercepts of the parabola $y = x^2 - 4x - 3$?
(A) $x = -1, x = 3$
(B) $x = -3, x = 1$
(C) $x = 1, x = -1$
(D) $x = 3, x = -3$
(E) $x = 0, x = 4$

Q6. A lab experiment involves a quadratic relationship between temperature and time: $y = t^2 - 2t - 1$. What is the minimum value of the function?
(A) $y = -2$
(B) $y = -1$
(C) $y = 0$
(D) $y = 1$
(E) $y = 2$

Q7. What is the equation of the axis of symmetry for the parabola $y = x^2 + 2x - 3$?
(A) $x = -1$
(B) $x = 1$
(C) $x = 2$
(D) $x = 3$
(E) $x = -2$

Q8. A physicist is studying the trajectory of a projectile: $y = -x^2 + 4x + 2$. What is the vertex of the parabola?
(A) $(0, 2)$
(B) $(2, 4)$
(C) $(2, 6)$
(D) $(4, 2)$
(E) $(4, 0)$

Open Ended Questions (FRQ)

Q9. Graph the parabola $y = x^2 - 2x - 3$. Identify the vertex, axis of symmetry, and x-intercepts.

Q10. A space station is moving with a velocity of $v = t^2 + 2t - 1$. Find the maximum value of the velocity and the time at which it occurs.

Chef's Tips

- Tip 1: Use graphing calculators to visualize the parabolas.
- Tip 2: Emphasize the importance of identifying the vertex and axis of symmetry.
- Tip 3: Encourage students to use real-world examples to illustrate the concepts.